

Status

Related asset, if applicable

16. Dashboard and Exception Requirements

Sprint 11 must add or populate exceptions for:

Inventory below configured threshold
Inventory reserved but not issued
Inventory issued but not installed
Missing serial number
Duplicate serial number
Inventory with missing cost layer
Inventory allocated to multiple projects
Negative inventory quantity
Damaged inventory awaiting disposition
Returned inventory awaiting inspection
Asset missing placed-in-service date
Asset missing useful life
Asset with depreciation exceeding depreciable basis
Asset with negative net book value
Asset past expected useful life
Asset awaiting disposal
Loan payment overdue
Loan balance mismatch
Loan missing payment allocation
Loan approaching maturity
Missing source payable or acquisition document

17. Sprint 11 Decisions

Decision 11-01 — FIFO inventory

Inventory is valued and issued using First In, First Out by product identity and cost layer.

Decision 11-02 — Serialized equipment identity

Serial-numbered equipment remains individually identifiable throughout its lifecycle.

Serial numbers cannot be reused after retirement, disposal, or cancellation.

Decision 11-03 — Inventory history

Inventory changes are recorded as transactions.

The system must not rely on manually overwriting a single quantity or value field.

Decision 11-04 — Project cost integration

Issued inventory must flow into the applicable project cost category:

Equipment Cost for equipment

Material Cost for materials

The source inventory item, serial number, cost layer, and project must remain linked.

Decision 11-05 — Historical project profitability

A later inventory or acquisition-cost correction must not silently recalculate a historical project's profitability.

The correction must create a formal project-cost adjustment linked to the original project and calculation snapshot.

Decision 11-06 — Depreciation method
All fixed assets use straight-line depreciation.

Accelerated and declining-balance depreciation are not supported in this sprint.

Decision 11-07 — Depreciation start date
Depreciation begins on the placed-in-service date, not the acquisition date.

Decision 11-08 — Depreciation preservation
Prior-period depreciation entries remain preserved.

Corrections create adjustment entries rather than rewriting historical depreciation.

Decision 11-09 — Asset categories
The initial fixed-asset categories are:

Vehicles

Computers

Offices

Additional categories may be created through controlled administration.

Decision 11-10 — Loan separation
Principal, interest, and fees are tracked separately for every loan payment.

Decision 11-11 — Manual loan entry
Loans and loan payments are manually recorded during the initial implementation.

Automatic lender or bank imports are deferred.

Decision 11-12 — Accounting integration
Inventory, asset, depreciation, and loan activities create ledger-compatible events, with final chart-of-accounts mapping completed during Sprint 12.

Decision 11-13 — Historical asset records
Fixed assets and loans cannot be silently deleted after financial activity exists.

Corrections require formal adjustment or cancellation records.

Decision 11-14 — Security
Inventory costs, asset values, depreciation, and loan balances are financial information and are permission-controlled.

18. Open Questions

Open Question 11-01 — Inventory valuation scope
Should FIFO be applied separately by:

Product identity

Brand and model

Product identity and location

Product identity, location, and condition

Recommended default: Product identity and cost layer, with location tracked separately.

Open Question 11-02 — Inventory locations

Will GECC have one inventory location or multiple locations?

Recommended default: Support multiple locations in the data model, even if only one location is configured initially.

Open Question 11-03 — Serialized material rules

Which product categories require serial-number tracking?

Recommended default: Make serialization configurable by product identity. Equipment requiring individual traceability must be serialized; consumable materials may remain quantity-based.

Open Question 11-04 — Inventory reservation expiration

How long may inventory remain reserved for a project before it is released?

Recommended default: No automatic release initially. Create an exception for stale reservations and require authorized review.

Open Question 11-05 — Depreciation frequency

Should depreciation be calculated monthly, daily, or by another convention?

Recommended default: Monthly depreciation with a documented month-end convention.

Open Question 11-06 — Partial-month depreciation

How should depreciation be handled when an asset is placed in service during a month?

Recommended default: Defer the final policy to Sprint 12 accounting configuration. The system must support the selected convention without rewriting prior entries.

Open Question 11-07 — Salvage value

Should fixed assets include salvage value?

Recommended default: Include an optional salvage-value field. Default to zero until an accounting policy is approved.

Open Question 11-08 — Asset impairment

Should impairment adjustments be supported?

Recommended default: Preserve an adjustment field and audit structure, but defer the impairment workflow until a later accounting decision.

Open Question 11-09 — Capitalization threshold

What minimum dollar amount requires a fixed-asset record rather than an expense record?

Recommended default: Make the threshold configurable and require the Comptroller to approve the initial value.

Open Question 11-10 — Useful lives

What useful lives apply to:

Vehicles

Computers

Offices

Additional asset categories

Recommended default: Store useful life per asset and require the Comptroller to approve category defaults before depreciation begins.

Open Question 11-11 — Loan amortization

Should the system calculate an amortization schedule or only record manually supplied principal and interest allocations?

Recommended default: Support manual payment allocation initially and design the model to support calculated schedules later.

Open Question 11-12 — Loan interest accrual

Should interest be accrued between payment dates?

Recommended default: Defer interest accrual until Sprint 12 unless required for the initial financial reports.

Open Question 11-13 — Asset disposal approval

Who must approve the disposal of a fixed asset?

Recommended default: Require Sales Manager or Comptroller approval, with the Comptroller responsible for financial review.

Open Question 11-14 — Inventory write-offs

Who may approve damaged, lost, or scrapped inventory?

Recommended default: Require Sales Manager or Comptroller approval and a documented reason.

19. Acceptance Criteria

Sprint 11 is accepted when:

Product identities can be created and maintained.

Serialized inventory items retain unique serial-number records.

Duplicate serial numbers are prevented.

Nonserialized materials can be tracked by quantity and cost layer.

Inventory receipts create cost layers.

FIFO issuance consumes the oldest available cost layer first.

Issuance across multiple cost layers calculates the correct total cost.

Serialized equipment remains linked to its individual serial number.

Inventory allocation is linked to the correct project and Master Sales Record version.

Inventory cannot be allocated to two active projects simultaneously.

Inventory returns require a condition or disposition decision.

Inventory adjustments preserve the original transaction.

Issued equipment and materials flow to the correct project-cost category.

Later inventory-cost corrections create formal adjustments rather than silently changing historical project profitability.

Vehicles, computers, and offices can be recorded as fixed assets.

Additional asset categories can be created through controlled administration.

Fixed assets retain acquisition cost, placed-in-service date, useful life, accumulated depreciation, and net book value.

Depreciation does not begin before the placed-in-service date.

Straight-line depreciation is calculated correctly.

Accumulated depreciation cannot exceed the depreciable basis without an approved adjustment.
Net book value is calculated correctly.
Historical depreciation entries cannot be silently overwritten.
Asset disposal preserves final book value and disposal information.
Loans can be recorded with lender, amount, date, term, rate, and payment schedule.
Loan payments separate principal, interest, and fees.
Principal balances recalculate correctly after payments.
Loan records cannot be deleted after financial activity exists.
Inventory, asset, depreciation, and loan reports are searchable and reproducible.
Unauthorized users cannot view restricted financial values.
Dashboard exceptions identify missing, inconsistent, or overdue records.
Inventory, asset, depreciation, and loan activities create accounting integration events.
All required actions are recorded in the tamper-evident audit log.
Automated Sprint 11 tests pass.

20. Automated Test Scenarios

Inventory Tests

Create a product identity.
Receive inventory from a vendor.
Create a cost layer.
Receive the same product at a different cost.
Issue inventory and confirm FIFO consumption.
Issue quantity across two cost layers.
Confirm remaining quantities in each layer.
Attempt to issue unavailable inventory.
Attempt to allocate one serialized item to two projects.
Attempt to reuse a retired serial number.
Return an issued item and require condition disposition.
Correct an inventory cost after project issuance.
Confirm the original inventory transaction remains unchanged.

Project-Cost Tests

Issue equipment to a project and confirm Equipment Cost integration.
Issue materials to a project and confirm Material Cost integration.
Confirm the Master Sales Record version is retained.
Confirm a later inventory correction creates a project-cost adjustment.
Confirm historical Sprint 10 profitability remains preserved.

Fixed-Asset Tests

Create a vehicle asset.
Create a computer asset.
Create an office asset.
Create an additional asset category.
Set acquisition date and placed-in-service date.
Attempt to calculate depreciation before placed-in-service date.
Calculate straight-line depreciation.
Confirm accumulated depreciation and net book value.
Attempt to exceed depreciable basis.
Dispose of an asset and preserve final values.
Confirm historical depreciation entries cannot be deleted.

Loan Tests

Create a loan.
Record a loan payment.
Split a payment between principal, interest, and fees.
Confirm the principal balance decreases by the principal portion only.
Mark a loan paid off only after the balance is resolved.

Correct a loan payment and preserve the original transaction.
Confirm loan activity creates accounting events.
Security and Audit Tests
Confirm a Technician cannot view acquisition costs.
Confirm an Accounts Payable Associate cannot change depreciation.
Confirm unauthorized users cannot edit loan balances.
Confirm asset-disposal approval is required.
Confirm inventory adjustments are audited.
Verify audit-chain integrity after corrections.
Confirm financial records cannot be silently deleted.

21. Sprint 11 Completion Definition

Sprint 11 is complete when:

Inventory is tracked by product identity, quantity, serial number, and FIFO cost layer.
Inventory is traceable from receipt through project allocation, installation, return, or disposal.
Project costs receive the correct inventory costs.
Fixed assets are recorded and depreciated using the approved straight-line method.
Asset disposal is controlled and auditable.
Loans and loan payments track principal, interest, and fees separately.
Reports reconcile to source transactions.
Accounting integration events are available for Sprint 12.
Permissions prevent unauthorized access or modification.
Dashboard exceptions are operational.
Audit records are complete and tamper-evident.
Acceptance tests pass.
Open questions are resolved or formally carried forward.
The Sprint 11 bundle is approved and preserved unchanged.

22. Next Markdown Bundle

The next bundle is:

text

GECC-12-financial-reporting.md

Its objective is to implement:

Double-entry financial ledger integration
Cash-basis financial reporting
Balance sheet
Cash-flow statement
Statement of operations
Accounts-receivable report
Accounts-payable report
Inventory report
Depreciation report
Loan report
Fixed-asset report
Retained earnings report
Working Capital Requirement report
Customer profitability report
Project profitability report
Monthly reporting periods
Fifth-calendar-day report preparation

Financial statement approval and signatures

Period publication

Period locking

Formal adjustment entries

Published-report preservation

Financial-report permissions

Financial-report audit logging

Reproducibility and report-integrity testing

===== END FILE: GECC-11-Inventory-Assets-And-Loans.md =====

=== BEGIN FILE: GECC-12-Financial-And-Reporting.md ===

Sprint 011 approval recorded.

GECC-12-financial-reporting.md

GECC Sales Back Office System — Sprint 12

Bundle Metadata

Bundle name: GECC-12-financial-reporting.md

Project: GECC Sales Back Office System

Company: Go Ecco Climate Control

Sprint: 12

Sprint name: Financial Ledger and Reporting

Baseline: GECC-00-project-baseline.md

Previous bundle: GECC-11-inventory-assets-and-loans.md

Status: Implementation bundle

Date: 2026-09-20

Next planned bundle: GECC-13-dashboard-and-search.md

Primary database: Relational database

System of record: GECC database

Google Sheets: Export or reporting destination only

Sprint Objective

Implement the internal double-entry accounting ledger, cash-basis financial reporting, monthly reporting periods, report approval and publication, period locking, formal adjustment entries, and required financial reports.

Sprint 12 must ensure that:

Financial events originate from controlled operational records.

Every posted ledger entry balances.

Financial reports are reproducible from source transactions.

Cash-basis revenue and expenses are reported consistently.

Balance-sheet accounts continue to reflect assets, liabilities, and equity.

Published periods are locked.

Corrections use formal adjustment entries.

Published financial statements remain permanently preserved.

The Sales Manager and Comptroller approve and sign published financial statements.

Financial reports remain auditable and permission-controlled.

2. Sprint Scope

2.1 Included

Chart of accounts

Double-entry journal

Journal entries

Journal-entry lines

Source-event references

Cash receipts

Cash disbursements

Accounts receivable

Accounts payable

Inventory
Fixed assets
Depreciation
Loans
Retained earnings
Financial-report periods
Period close
Period publication
Period locking
Formal adjustments
Balance sheet
Cash-flow statement
Statement of operations
Accounts-receivable report
Accounts-payable report
Inventory report
Depreciation report
Loan report
Fixed-asset report
Retained-earnings report
Working Capital Requirement report
Customer profitability report
Project profitability report
Financial signature workflow
Financial-report audit logging
Report exports and preservation
2.2 Excluded
Automatic bank-statement import
Automatic bank reconciliation
Automatic transaction matching
Tax-return preparation
Tax-basis depreciation
Payroll tax calculation
Contractor tax reporting
External audit management
Budgeting and forecasting
Consolidated reporting for multiple legal entities
Automatic lender integration
3. Financial Architecture
3.1 Ledger as Internal Accounting Foundation
The system must use a double-entry ledger internally.
Each posted journal entry must satisfy:
text

Total Debits = Total Credits

Operational records remain the source of business events. The ledger records the accounting effect of those events.

Examples of source events include:

Verified customer payment
Approved customer refund
Vendor payment
Payable creation
Inventory receipt

Inventory issuance
Fixed-asset acquisition
Depreciation
Loan origination
Loan payment
Retained-earnings closing entry

Formal period adjustment

3.2 Source-Event Traceability

Every journal entry must reference:

Source event type

Source record ID

Project, when applicable

Customer, when applicable

Vendor, when applicable

Reporting period

User or system process creating the entry

Creation date and time

Reversal or adjustment reference, if applicable

A journal entry must not be posted without a valid source event or an authorized formal adjustment.

3.3 Posted Entries

Posted journal entries are immutable.

Corrections require:

Reversing entry

Replacement entry

Formal adjustment entry

Reason

Authorized user

Audit record

Link to the original entry

The system must not permit ordinary users to edit or delete posted entries.

Chart of Accounts

The system must support a configurable chart of accounts with at least these account classes:

Assets

Liabilities

Equity

Revenue

Expenses

Contra accounts

Cash-flow classification references

4.1 Initial Account Groups

The initial configuration must support accounts for:

Assets

Cash

Accounts receivable

Inventory

Fixed assets

Accumulated depreciation

Other approved assets

Liabilities

Accounts payable

Loans payable

Other approved liabilities

Equity
Owner or company equity
Retained earnings
Current-period earnings, if used
Revenue
Cash-basis project revenue
Other approved revenue categories
Expenses
Equipment cost
Material cost
Dealer fees
Lead cost
Technician cost
Damon cost
Jerry cost
Contractor commissions
Depreciation expense
Interest expense
Overhead expenses
Other approved expense categories

The exact account numbers and final account names are configurable.

4.2 Account Controls

Each account must include:

Account ID
Account number
Account name
Account class
Normal balance
Active or inactive status
Report classification
Cash-flow classification
Parent account, if applicable
Effective date
Deactivation date
Description

An account used by a posted journal entry may not be deleted.

Journal Data Model

5.1 Journal Entry

Required fields:

Journal entry ID
Journal-entry number
Entry date
Posting date
Reporting period
Entry type
Source event type
Source record ID
Description
Total debits
Total credits
Status
Created by
Approved by, if required

Created date and time

Posted date and time

Reversal reference

Adjustment reference

Notes

5.2 Journal Entry Line

Required fields:

Journal-entry line ID

Journal entry ID

Account ID

Debit amount

Credit amount

Customer reference, if applicable

Project reference, if applicable

Vendor reference, if applicable

Contractor reference, if applicable

Cost category, if applicable

Description

Source reference

A journal line must have either a debit or credit amount, not both.

5.3 Entry Statuses

Permitted statuses:

DRAFT

PENDING_REVIEW

APPROVED

POSTED

REVERSED

VOIDED

A POSTED entry cannot return to DRAFT.

Required Accounting Events

6.1 Verified Customer Payment

A verified customer payment must create a cash-receipt accounting event.

The event must reference:

Payment

Invoice

Project

Customer

Verification event

Payment date

Reporting period

The default accounting structure is:

Debit: Cash

Credit: Accounts receivable or cash-basis revenue account, according to the configured revenue-recognition policy

The final account mapping must be configurable without changing the source payment record.

6.2 Customer Refund

An approved refund must create a cash-disbursement or receivable-reduction event.

The event must reference:

Original payment or invoice

Refund approval

Project

Customer

Refund date

Amount

6.3 Vendor Payable Creation

When the Certificate is fully signed and required payables are created, the system must create the applicable payable obligation.

The event must reference:

Payable

Project

Vendor

Certificate

Payable category

Certificate signing date

6.4 Vendor Payment

A payable payment must create a cash-disbursement event referencing:

Payable

Vendor

Project

Payment date

Payment method

Amount

The accounting event must reduce cash and the applicable payable balance.

6.5 Inventory

The ledger must support events for:

Inventory receipt

Inventory issue

Inventory return

Inventory adjustment

Inventory write-off, if approved

6.6 Fixed Assets

The ledger must support events for:

Fixed-asset acquisition

Asset placed in service

Depreciation

Asset disposal

Asset-value adjustment

6.7 Loans

The ledger must support events for:

Loan origination

Principal payment

Interest payment

Loan fee payment

Principal adjustment

Loan payoff

Principal and interest must remain separately identifiable.

Cash-Basis Reporting

Financial reporting is cash-basis, subject to the final accounting policy approved by the Comptroller.

7.1 Revenue

Revenue is recognized in cash-basis reports only when a customer payment is verified and recognized as received.

An invoice or accounts-receivable balance alone does not create cash-basis revenue.

7.2 Expenses

Cash-basis operating expenses are recognized when the related cash disbursement is recorded and posted.

The system must distinguish between:

Expense obligation

Payable creation

Cash disbursement

Cash-basis expense recognition

7.3 Balance-Sheet Accounts

The balance sheet must continue to report relevant non-cash balances, including:

Cash

Accounts receivable

Inventory

Fixed assets

Accumulated depreciation

Accounts payable

Loans payable

Equity

Retained earnings

The system must not remove accounts payable merely because the primary statement of operations uses cash-basis recognition.

7.4 Non-Cash Events

Non-cash events, including depreciation and payable creation, must remain in the ledger and appear in the applicable supporting reports even when excluded from cash-basis operating results.

Financial-Report Periods

8.1 Period Fields

Each financial-report period must include:

Period ID

Month

Year

Period start date

Period end date

Report preparation date

Report publication date

Approval status

Signature status

Locked status

Adjustment references

Created date and time

Closed date and time

Published by

8.2 Monthly Reporting Date

Monthly reports are prepared on the fifth calendar day of the following month.

Example:

text

May reports are prepared on June 5.

The report preparation date must be stored even if preparation occurs later.

8.3 Period Statuses

Permitted statuses:

OPEN

PREPARING

PENDING_APPROVAL

APPROVED

PUBLISHED

LOCKED

ADJUSTMENT_PENDING

REOPENED_BY_AUTHORIZED_EXCEPTION

8.4 Period Locking

Once financial statements are published:

The period becomes locked.

Ordinary edits are prohibited.

Posted entries cannot be changed.

Corrections require formal adjustment entries.

Original published statements remain preserved.

Any adjustment must reference the locked period.

A locked period may not be reopened through ordinary user actions.

Financial Statement Workflow

Step 1 — Period preparation

The system identifies the prior month and prepares the applicable reports on the fifth calendar day of the following month.

Step 2 — Source reconciliation

The system validates:

Journal-entry balance

Payment totals

Accounts-receivable totals

Accounts-payable totals

Inventory totals

Fixed-asset totals

Depreciation totals

Loan balances

Retained earnings

Report-to-ledger consistency

Step 3 — Review

The Sales Manager and Comptroller review the reports and exceptions.

Step 4 — Corrections

Errors must be corrected through:

Unposted-entry correction, if the period remains open

Formal adjustment entry, if the period is published or locked

Step 5 — Approval and signatures

Published financial statements must be approved and signed by:

Sales Manager

Comptroller

The Installation Manager does not sign financial statements.

Step 6 — Publication

After required approval and signatures:

Reports receive a publication date.

Report versions are finalized.

Reports are retained permanently.

The period is locked.

Distribution events are recorded.

10. Required Financial Reports

10.1 Balance Sheet

The balance sheet must report, by period:

Assets

Liabilities

Equity

Retained earnings

Total assets

Total liabilities and equity

The system must validate:
text

Total Assets = Total Liabilities + Total Equity

Any imbalance must block publication.

10.2 Cash-Flow Statement

The cash-flow statement must report:

Beginning cash

Customer cash receipts

Vendor cash disbursements

Contractor or commission cash disbursements

Overhead cash disbursements

Loan proceeds

Loan principal payments

Loan interest payments

Asset purchases

Asset-disposal proceeds

Other approved cash movements

Net change in cash

Ending cash

Ending cash must reconcile to the applicable cash-ledger balance.

10.3 Statement of Operations

The statement of operations must report cash-basis results, including:

Cash-basis revenue

Cash-basis project costs

Cash-basis contractor commissions

Cash-basis overhead expenses

Depreciation, according to the approved reporting presentation

Interest expense, according to the approved reporting presentation

Net income or loss

The report must clearly identify any non-cash items included for management or financial-reporting purposes.

10.4 Accounts-Receiveable Report

The report must include:

Customer

Project

Invoice

Original invoice amount

Approved adjustments

Verified payments

Remaining balance

Payment status

Verification status

Aging information

10.5 Accounts-Payable Report

The report must include:

Vendor

Project

Payable category

Amount owed

Amount paid

Outstanding balance

Certificate signing date

Due date

Payment status

Approval status

10.6 Inventory Report

The report must include:

Product identity

Serial number, if applicable

Quantity

Inventory status

Cost layer

Unit cost

Total inventory value

Project allocation

Available quantity

10.7 Depreciation Report

The report must include:

Asset

Category

Acquisition cost

Placed-in-service date

Useful life

Depreciation method

Current-period depreciation

Accumulated depreciation

Net book value

Adjustment references

10.8 Loan Report

The report must include:

Lender

Loan amount

Loan date

Interest rate

Term

Principal balance

Principal paid

Interest paid

Fees

Payment schedule

Maturity date

Loan status

10.9 Fixed-Asset Report

The report must include:

Asset ID

Asset category

Description

Acquisition date

Acquisition cost

Placed-in-service date

Accumulated depreciation

Net book value

Location

Disposal status

10.10 Retained-Earnings Report

The report must include:

Beginning retained earnings

Prior-period adjustments

Current-period net income or loss

Distributions, if applicable

Other approved equity adjustments

Ending retained earnings

10.11 Working Capital Requirement Report

The report must include configurable working-capital components, initially including:

Accounts receivable

Inventory

Other current assets

Accounts payable

Other current liabilities

Working capital

Cash available

Required working capital

Working-capital surplus or deficiency

The final formula and required operating assumptions must be approved by the Comptroller.

10.12 Customer Profitability Report

The report must aggregate authorized project information by customer, including:

Customer

Project count

Invoice amounts

Verified cash receipts

Project costs

Commissions

Adjustments

Gross Profit

Net Profit

10.13 Project Profitability Report

The report must include:

Project identification code

Customer

Sales Associate

Invoice amount

Cost categories

Gross Profit

Variable commissions

Fixed compensation

Net Profit

Payment status

Project completion status

Adjustment references

11. Financial Signatures and Distribution

11.1 Required Signers

Published financial statements require:

Sales Manager approval and signature

Comptroller approval and signature

The Installation Manager does not sign financial statements.

11.2 Signature Records

Each signature must retain:

Signer identity

Signer role

Signature date and time

Report version

Period

Signature status

Signature certificate or equivalent audit reference

11.3 Report Distribution

The system must record:

Report recipients

Delivery method

Delivery date and time

Delivery status

Failure reason, if applicable

Report version distributed

Published reports must remain downloadable by authorized users.

Formal Adjustments

A formal adjustment is required when:

A published period contains an error.

A locked period requires correction.

A prior-period cost is corrected.

A prior-period payment is reversed.

A prior-period depreciation value is corrected.

A prior-period loan balance is corrected.

A prior-period inventory value is corrected.

A retained-earnings correction is required.

Each adjustment must include:

Adjustment ID

Affected period

Adjustment date

Effective period

Debit and credit accounts

Amount

Reason

Supporting source records

Requested by

Approved by

Approval date and time

Related original journal entries

Related reports

Notes

The original published statement must remain unchanged.

Permissions

Sales Manager

May:

View all financial reports.

Review financial statements.

Approve and sign published financial statements.

Review period exceptions.

Approve permitted adjustments.

View audit history.

Comptroller

May:

View all financial reports.

Review and approve financial statements.

Sign published financial statements.

Manage financial reporting.

Approve permitted adjustments.

Review ledger integrity.

View audit history.

Installation Manager

May view financial information only as authorized for operational duties.

The Installation Manager may not:

Sign financial statements.

Publish financial statements.

Unlock a closed period.

Approve retained-earnings adjustments.

Accounts Payable Associate

May view authorized payable and disbursement reports.

May not:

Publish financial statements.

Post arbitrary journal entries.

Unlock periods.

Approve financial statements.

Database Administrator

May maintain authorized database and reporting functions.

May not automatically:

Approve financial statements.

Publish reports.

Alter posted journal entries.

Delete financial records.

Sales Associates and Contractors

May access only authorized project profitability or commission information.

They may not access:

General financial statements

Other Sales Associates' commission data

Retained earnings

Loan reports

Company-wide cash balances

Unauthorized accounts payable or receivable totals

14. Audit Requirements

The system must log:

Chart-of-accounts creation and changes

Journal-entry creation

Journal-entry approval

Journal-entry posting

Journal-entry reversal

Formal adjustment creation

Formal adjustment approval

Period preparation

Report generation

Report recalculation

Report approval

Report signature

Report publication

Period locking
Authorized exception reopening
Report export
Report download
Report print
Report distribution
Failed report delivery
Unauthorized financial access
Failed ledger-integrity checks
All entries must use the tamper-evident audit chain.

Sprint 12 Decisions

Decision 12-01 — Internal double-entry ledger

The system will maintain a double-entry ledger as the internal accounting foundation.

Decision 12-02 — Balanced journal entries

No journal entry may be posted unless total debits equal total credits.

Decision 12-03 — Source-event accounting

Operational transactions create accounting events. Users may not bypass source records to create unsupported financial activity except through an authorized formal-adjustment workflow.

Decision 12-04 — Cash-basis reporting

Revenue and expenses in the primary operating reports follow the approved cash-basis policy.

Verified customer receipts are required for cash-basis revenue recognition.

Cash disbursements are required for cash-basis expense recognition, subject to the approved presentation of non-cash items.

Decision 12-05 — Balance-sheet continuity

Accounts receivable, accounts payable, inventory, fixed assets, accumulated depreciation, loans, and equity remain available for balance-sheet reporting even when operating results are reported on a cash basis.

Decision 12-06 — Monthly reporting

Financial reporting periods are monthly.

Reports are prepared on the fifth calendar day of the following month.

Decision 12-07 — Published-period locking

Published periods are locked.

Ordinary edits are prohibited after publication.

Decision 12-08 — Formal correction process

Corrections to published or locked periods require formal adjustment entries.

Original entries and published statements remain preserved.

Decision 12-09 — Required financial signers

The Sales Manager and Comptroller approve and sign published financial statements.

The Installation Manager does not sign financial statements.

Decision 12-10 — Ledger immutability

Posted journal entries cannot be edited or deleted through the application.

Corrections use reversal, replacement, or formal adjustment entries.

Decision 12-11 — Report reproducibility

Reports must be reproducible from the ledger, source records, report parameters, and approved report version.

Decision 12-12 — Financial-report access

Financial reports and ledger details are permission-controlled and cannot be exposed to unauthorized users through search, export, dashboard, or document links.

Decision 12-13 — Report preservation

Published financial reports, signatures, approval records, and distribution records are retained permanently.

Open Questions

Open Question 12-01 — Cash-basis statement of operations

Should depreciation appear in the primary cash-basis statement of operations or only in a supporting management presentation?

Recommended default: Include depreciation as a separately identified non-cash line when required for management reporting, while also presenting a clearly labeled cash-basis operating result.

Open Question 12-02 — Accounts-payable treatment

How should Certificate-triggered accounts payable appear in the cash-basis financial statements before vendor payment?

Recommended default: Include the payable on the balance sheet and supporting accounts-payable report, but recognize the expense in the cash-basis operating statement when paid unless the Comptroller approves another treatment.

Open Question 12-03 — Revenue amount after adjustments

Should cash-basis revenue be based on:

Gross verified customer receipts

Adjusted invoice amount

Verified receipts net of refunds

Another approved treatment

Recommended default: Use verified receipts net of approved refunds for cash-basis revenue, while preserving invoice and receivable values separately.

Open Question 12-04 — Period-end cutoff

What time zone and cutoff time determine the reporting period for payments, disbursements, and adjustments?

Recommended default: Use the company's configured operating time zone and 11:59:59 p.m. local time on the last calendar day of the period.

Open Question 12-05 — Report preparation on non-business days

If the fifth calendar day is a weekend or holiday, should reports be prepared:

On the fifth calendar day

On the preceding business day

On the next business day

Recommended default: Preserve the fifth calendar day as the report preparation date and permit actual preparation on the next business day.

Open Question 12-06 — Retained-earnings closing

Should net income or loss close to retained earnings monthly or annually?

Recommended default: Track current-period earnings monthly and perform formal retained-earnings closing annually unless the Comptroller requires monthly closing.

Open Question 12-07 — Working Capital Requirement formula

Which operational assumptions should determine required working capital?

Recommended default: Make the formula configurable and require Comptroller approval before publication of the first official Working Capital Requirement report.

Open Question 12-08 — Journal-entry approval

Which journal entries require separate review before posting?

Recommended default: Automatically generated entries may post after validation; manual and formal-adjustment entries require review by the Comptroller or another authorized approver.

Open Question 12-09 — Report version naming

Should published reports use sequential version numbers, hash-based identifiers, or both?

Recommended default: Use a human-readable sequential report version plus an integrity hash.

Open Question 12-10 — Period reopening

Can a published period ever be reopened, or must every correction use a later-period adjustment?

Recommended default: Do not reopen published periods through ordinary workflows. Use formal adjustments in the current open period.

Open Question 12-11 — Cash-flow classification

How should loan proceeds, principal payments, interest, asset purchases, and asset-disposal proceeds be classified in the cash-flow statement?

Recommended default: Configure classifications by account and event type, subject to Comptroller approval.

Open Question 12-12 — Financial statement distribution

Who receives published financial statements and supporting reports?

Recommended default: Configure a distribution list limited to the Sales Manager, Comptroller, and explicitly authorized recipients.

Acceptance Criteria

Sprint 12 is accepted when:

A chart of accounts can be configured.

Journal entries contain balanced debit and credit lines.

Unbalanced journal entries cannot be posted.

Every posted entry references a valid source event or authorized formal adjustment.

Posted journal entries cannot be silently edited or deleted.

Verified customer payments create accounting events.

Vendor payments create accounting events.

Inventory events create accounting events.

Fixed-asset acquisitions create accounting events.

Depreciation creates accounting events.

Loan origination and payments create accounting events.

Accounts receivable and accounts payable reconcile to source records.

Monthly financial periods are created correctly.

Reports identify the fifth calendar day of the following month as the preparation date.

Balance sheets balance.

Ending cash on the cash-flow statement reconciles to the ledger cash balance.

Cash-basis revenue excludes unverified customer payments.

Cash-basis expenses follow the approved cash-disbursement policy.

Published financial statements require Sales Manager and Comptroller approval and signatures.

The Installation Manager cannot sign published financial statements.

Published reports receive immutable report versions.

Published periods become locked.

Ordinary edits are blocked after publication.

Corrections to locked periods require formal adjustment entries.

Original published statements remain preserved after adjustment.

Retained earnings are calculated from approved source balances.

Working Capital Requirement reports use a configured and approved formula.

Project and customer profitability reports reconcile to Sprint 10 records.

Inventory, asset, depreciation, loan, receivable, and payable reports reconcile to source records.

Unauthorized users cannot view or export restricted financial information.

Report generation, approval, signing, publication, locking, and adjustment actions are audited.

Financial reports are reproducible using stored report parameters and versions.

Automated Sprint 12 tests pass.

18. Automated Test Scenarios

Ledger Tests

Create a balanced journal entry.

Reject an unbalanced journal entry.

Reject a journal entry without a source reference.

Post a verified-payment entry.

Post a vendor-payment entry.

Reverse a posted entry.

Confirm the original entry remains unchanged.

Attempt to delete a posted entry.

Confirm deletion is blocked and audited.

Period Tests

Create a monthly reporting period.

Confirm the preparation date is the fifth calendar day of the following month.

Generate reports for an open period.

Publish reports after required approvals and signatures.

Confirm publication locks the period.

Attempt an ordinary edit after locking.

Create a formal adjustment against a locked period.

Confirm the original report remains preserved.

Statement Tests

Confirm the balance sheet balances.

Confirm ending cash reconciles to the ledger.

Confirm unverified payments are excluded from cash revenue.

Confirm verified refunds reduce the applicable cash-basis result.

Confirm accounts payable remains visible on the balance sheet before payment.

Confirm depreciation is reported according to the configured presentation.

Confirm retained earnings reconcile.

Confirm Working Capital Requirement calculations use approved configuration.

Security Tests

Confirm Sales Associates cannot view company-wide financial statements.

Confirm Contractors cannot view general-ledger details.

Confirm Accounts Payable Associates cannot publish financial statements.

Confirm the Installation Manager cannot sign financial statements.

Confirm unauthorized exports are blocked and audited.

Confirm financial report access is recorded in the audit chain.

Reproducibility Tests

Generate a report twice using the same period and parameters.

Confirm identical underlying results.

Change a source record in an open period and confirm the report version changes appropriately.

Publish a report and confirm later source changes do not alter the published version.

Create an adjustment and confirm the original and corrected versions are both available.

19. Sprint 12 Completion Definition

Sprint 12 is complete when:

The double-entry ledger is operational.

Operational source events create traceable accounting entries.

Required financial reports are generated.

Cash-basis reporting rules are implemented and documented.

Monthly reporting periods are prepared, approved, published, and locked.

Formal adjustments correct published periods without overwriting history.

Sales Manager and Comptroller signatures are captured.

Reports are reproducible and permanently preserved.

Financial permissions are enforced.

Audit records are complete and tamper-evident.

Acceptance tests pass.

Open questions are resolved or formally carried forward.

The Sprint 12 bundle is approved and preserved unchanged.

20. Next Markdown Bundle

The next bundle is:

text

GECC-13-dashboard-and-search.md

Its objective is to implement:

Activity dashboard
Role-specific dashboard views
Project-status visibility
Current-task visibility
Next-required-action visibility
Due dates
Days in current status
Last-activity tracking
Warning and exception indicators
Global project search
Customer search
Contact search
Location search
Serial-number search
Equipment search
Vendor and payable search
Document search
Date-range filters
Payment and signature filters
Negative-profitability filters
Overdue-task filters
Expiring-document filters
Role-based search results
Financial-information masking
Dashboard and search audit logging

Search-performance and access-control testing

===== END FILE: GECC-12-Financial-And-Reporting.md =====

=== BEGIN FILE: GECC-13-Dashboard-And-Search.md ===

GECC-13-dashboard-and-search.md

GECC Sales Back Office System — Sprint 13

Bundle Metadata

Bundle name: GECC-13-dashboard-and-search.md

Project: GECC Sales Back Office System

Company: Go Ecco Climate Control

Sprint: 13

Sprint name: Activity Dashboard and Search

Baseline: GECC-00-project-baseline.md

Previous bundle: GECC-12-financial-reporting.md

Status: Implementation bundle

Date: 2026-09-20

Next planned bundle: GECC-14-notifications-and-files.md

Primary database: Relational database

System of record: GECC database

Google Sheets: Export or reporting destination only

Sprint Objective

Implement the activity dashboard, role-specific dashboard views, authorized global search, operational filters, exception indicators, and related audit logging.

Sprint 13 must ensure that authorized users can quickly identify:

Current project status

Current task

Assigned user or role

Next required action

Overdue work

Expiring documents
Payment and signature exceptions
Installation status
Negative-profit projects
Commission and financial exceptions permitted by their role
Search must provide useful access to authorized records without exposing restricted records, financial information, or the existence of unauthorized data.
Sprint Scope
2.1 Included
Activity dashboard
Role-specific dashboard views
Project status summaries
Current-task indicators
Next-action indicators
Assignment indicators
Due dates
Days in current status
Last-activity date and time
Warning indicators
Exception indicators
Customer search
Contact search
Project search
Location search
Project-code search
Equipment search
Serial-number search
Vendor search
Payable search
Document search
Date-range filters
Status filters
Signature filters
Payment filters
State and location filters
Sales Associate and Technician filters
Overdue-task filters
Expiring-document filters
Negative-profit filters
Role-based result filtering
Financial-information masking
Dashboard and search audit logging
Search and dashboard performance testing
2.2 Excluded
Email and notification delivery
Customer-file document storage implementation
Electronic-signature implementation
New workflow states
New accounting calculations
Technician scheduling beyond existing installation-task records
GPS tracking
Technician time tracking
Unauthorized cross-role data access

Dashboard Design

3.1 Dashboard Record

The dashboard must be generated from current authoritative records rather than manually maintained status fields.

Each project dashboard item must be capable of displaying:

Project identification code

Customer name

Project location

Assigned Sales Associate

Assigned Technician or Technicians

Current project status

Current task

Assigned role or user

Contract status

Invoice status

Certificate status

Installation status

Payment status

Commission status

Next required action

Due date

Days in current status

Last activity date and time

Warning indicator

Exception indicator

3.2 Dashboard Data Freshness

Dashboard values must reflect current approved system records.

The implementation must define whether values are:

Calculated on request

Updated by event-driven projection

Refreshed on a scheduled interval

Refreshed manually

The dashboard must display the last refresh timestamp where data is not calculated synchronously.

3.3 Current Task

The current task is the next incomplete task permitted by the project state machine.

Examples include:

Review Master Sales Record

Approve Master Sales Record

Review generated contract

Obtain customer contract signature

Obtain GECC contract signature

Obtain GECC invoice signature

Obtain customer invoice signature

Schedule installation

Wait for installation eligibility date

Complete scope of work

Obtain customer certificate signature

Obtain Installation Manager certificate signature

Verify payment

Approve adjustment

Review commission report

The system must not display a task as complete when the underlying required action remains incomplete.

3.4 Next Required Action

The next required action must be derived from:

Current project status

Incomplete lifecycle requirement

Assigned role or user

Due date

Exception state

Required approval

Required signature

Required payment verification

Where multiple actions are available, the state-machine priority order must determine the primary displayed action.

Role-Specific Dashboard Views

4.1 Sales Associate

May view:

Assigned projects

Customer and project information

Contract status

Invoice status

Certificate status

Installation status

Payment status as operationally necessary

Current task

Next required action

Due dates

Signature exceptions

Commission status for their own sales

May not view:

Another Sales Associate's commission information

Unauthorized costs

Company-wide financial statements

Unauthorized payment amounts

Unauthorized payroll data

4.2 Sales Manager

May view:

All projects

All operational tasks

All financial exceptions

All payment and receivable exceptions

All commission exceptions

Negative-profit projects

Audit-related warnings

All authorized dashboard filters

4.3 Comptroller

May view:

All projects

All financial and accounting exceptions

Payment, receivable, payable, commission, and profitability information

Financial-report status

Period-lock and adjustment exceptions

All authorized dashboard filters

4.4 Installation Manager

May view:

Installation-related projects

Assigned technicians

Installation tasks

Scope status

Equipment and serial-number requirements

Installation-related costs when authorized

Certificate-signature tasks

Three-day waiting-period exceptions

May not view unauthorized commission, payroll, or financial-report information.

4.5 Technician

May view:

Assigned projects

Approved scope

Assigned tasks

Equipment information required for assigned work

Serial-number requirements

Installation status

Completion notes

Proposed scope-change tasks

May not view:

Unauthorized prices

Unauthorized costs

Commission information

Payroll information

Financial statements

Unassigned project details

4.6 Accounts Payable Associate

May view:

Authorized vendor records

Authorized payables

Payable statuses

Payment tasks

Project information required for payable duties

May not view unrelated project profitability, commissions, or financial statements.

4.7 Database Administrator

May view authorized operational and data-quality records.

The role may view data-quality exceptions but may not automatically receive financial approval authority.

4.8 Other Contractors

May view only assigned projects, assigned tasks, and required fields.

Dashboard Filters

The dashboard must support the following filters, subject to role authorization:

Project status

Customer

Project identification code

Street address

City

State

ZIP code

Sales Associate

Technician

Assigned role
Date range
Payment status
Contract status
Invoice status
Certificate status
Installation status
Signature status
Commission status
Overdue tasks
Expiring documents
Negative-profit projects
Projects requiring action
Current task
Exception type
Filters must be combinable.

For example:

Projects assigned to a Sales Associate with unsigned contracts

Projects in a particular state awaiting installation

Negative-profit completed projects

Unverified bank wires for a date range

Payables overdue for a selected vendor

Projects awaiting Certificate signature

A user must not be able to use a filter to reveal records that are otherwise unauthorized.

Exception Indicators

6.1 Signature Exceptions

The dashboard must identify:

Contracts awaiting customer signature

Contracts awaiting GECC signature

Invoices awaiting GECC signature

Invoices awaiting customer signature

Certificates awaiting customer signature

Certificates awaiting Installation Manager signature

Certificates awaiting Comptroller substitute signature

Signature requests nearing expiration

Expired signature requests

6.2 Installation Exceptions

The dashboard must identify:

Installations awaiting scheduling

Installation dates violating the three-day rule

Projects eligible for installation but not scheduled

Projects with incomplete installation records

Projects with completed work but missing certificate signatures

6.3 Payment Exceptions

The dashboard must identify:

Projects with completed work but no verified payment

Unpaid invoices

Partially paid invoices

Unverified cashier's checks

Unverified bank wires

Bank wires missing a reference number or proof

Overpayments

Payment reversals affecting completed projects

6.4 Scope and Data Exceptions

The dashboard must identify:

Scope changes awaiting approval

Missing equipment serial numbers

Missing required costs

Duplicate project-location warnings

Inconsistent project or invoice data

Missing vendor information

Invalid or incomplete document data

6.5 Profitability and Commission Exceptions

Authorized users may see:

Negative-profit projects

Zero-profit projects

Projects awaiting profitability calculation

Commission adjustments awaiting approval

Commission reports awaiting review

Commission reports awaiting approval

Post-payroll corrections

Financial amounts must remain hidden from users without financial authorization.

6.6 Reporting Exceptions

Authorized users may see:

Financial reports awaiting approval

Financial reports awaiting signatures

Closed-period adjustment requests

Ledger-balance errors

Unreconciled accounts receivable

Unreconciled accounts payable

Inventory or asset reconciliation exceptions

7. Search Requirements

7.1 Searchable Fields

Authorized users must be able to search by:

Customer name

Phone number

Email address

Project identification code

Street address

City

State

ZIP code

Sales Associate

Technician

Serial number

Brand

Model

Project status

Contract status

Invoice status

Certificate status

Payment status

Vendor

Payable

Document type

Date range

7.2 Search Behavior

Search must support:

- Exact matching
- Partial text matching
- Case-insensitive matching
- Normalized phone-number matching
- Normalized address matching
- Project-code matching
- Serial-number matching
- Filter combinations
- Sorting by relevance or date
- Pagination
- Empty-result handling
- Clear filters
- Saved searches, if authorized

The system must prevent unauthorized users from discovering restricted records through:

- Search results
- Result counts
- Autocomplete
- Error messages
- Duplicate warnings
- Export functions
- Search suggestions

7.3 Search Result Fields

Search-result fields must be role-specific.

Operational users may see:

- Project identification code
- Customer name
- Project location
- Assigned personnel
- Current project status
- Current task
- Document statuses
- Installation status

Payment status when authorized

Financial users may additionally see:

- Invoice amount
- Paid amount
- Remaining balance
- Profitability
- Costs
- Commission status
- Payable status

Unauthorized financial values must be masked or omitted, not displayed as blank values that imply existence.

7.4 Search Result Actions

Actions available from search results must be permission-controlled, including:

- Open customer
- Open project
- Open Master Sales Record
- Open document
- Open payment
- Open payable

Open commission record

Open financial report

Export results

Print results

Download document

8. Search and Dashboard Security

The system must apply authorization at:

Module level

Record level

Field level

Action level

Export level

Document level

A search query must execute against the user's authorized data scope.

The system must not:

Return unauthorized records

Reveal restricted record counts

Reveal restricted customer existence

Reveal restricted financial totals

Permit unauthorized sorting by hidden financial fields

Permit unauthorized export of hidden fields

Include restricted data in autocomplete suggestions

Every permission failure must be logged.

Dashboard and Search Audit Requirements

The system must log:

Dashboard access

Dashboard filter use

Dashboard result viewing

Search execution

Search filters

Search result viewing

Search result export

Search result printing

Search result downloads

Opening a record from search

Opening a document from search

Permission failures

Attempts to search restricted fields

Attempts to access restricted records

Saved-search creation or modification

Dashboard configuration changes

Each event must include:

User identity

Role

Date and time

Activity type

Query or filter reference, where appropriate

Related record identifier, where applicable

Success or failure

Session identifier

Reason when required

Previous audit-log entry reference

Integrity hash or equivalent value

Sensitive search terms must be handled according to the system's approved logging policy.

Performance and Usability Requirements

The system should:

Return ordinary dashboard views within the configured performance target.

Return ordinary searches within the configured performance target.

Use pagination for large result sets.

Avoid loading all records into the browser.

Provide a visible loading state.

Provide a clear no-results state.

Preserve filters when returning from a record.

Provide a clear way to reset filters.

Support keyboard-accessible controls.

Use consistent status and warning terminology.

Display dates and times in the configured company time zone.

Exact performance targets must be approved before production deployment.

Data and Search Indexing

The implementation may use relational indexes or a dedicated search index, provided that:

The database remains the authoritative source of truth.

Search indexes do not retain unauthorized data longer than permitted.

Index updates are auditable where required.

Deleted, cancelled, or superseded records are correctly represented.

Permissions are applied at query time.

Search results identify the current record version.

Historical document versions remain discoverable only to authorized users.

Search indexes must be refreshed when changes occur to:

Customer records

Project records

Assignments

Project status

Document status

Payment status

Equipment or serial numbers

Vendor and payable records

Commission status

Financial-report status

12. Sprint 13 Decisions

Decision 13-01 — Authoritative dashboard data

The dashboard is generated from current authoritative records and workflow state, not manually edited dashboard fields.

Decision 13-02 — Current task derivation

Current task is derived from the controlled lifecycle and the next incomplete required action.

Decision 13-03 — Role-specific views

Dashboard content, filters, search fields, result fields, and actions are role-specific.

Decision 13-04 — Search authorization

Search is constrained by the same record, field, and action permissions used elsewhere in the system.

Decision 13-05 — No unauthorized existence disclosure

The system must not reveal the existence, count, or identifying information of records outside the user's authorization scope.

Decision 13-06 — Financial masking

Financial amounts, commissions, profitability, payroll information, and ledger data are hidden or masked unless authorized.

Decision 13-07 — Exception-first dashboard

The dashboard prioritizes projects requiring action, overdue tasks, expiring documents, payment exceptions, signature exceptions, and data-integrity exceptions.

Decision 13-08 — Combined filters

Dashboard and search filters may be combined, but every filter combination remains subject to authorization.

Decision 13-09 — Project version display

The dashboard and search results display the current approved project or Master Sales Record version, while authorized users may access historical versions through the record.

Decision 13-10 — Search result pagination

Search results use pagination or controlled incremental loading to prevent large-result performance and data-exposure problems.

Decision 13-11 — Audited access

Dashboard access, searches, filter use, record openings, exports, downloads, and permission failures are audit logged.

Decision 13-12 — Search index authority

Any search index is a derived copy. The relational database remains the system of record.

Decision 13-13 — Status terminology

Dashboard and search status labels must use controlled enumerated values from the workflow and financial systems.

Decision 13-14 — Date and time display

Dashboard due dates, activity dates, and search date filters use the company's configured time zone.

Open Questions

Open Question 13-01 — Dashboard refresh model

Should dashboard data refresh:

On every page request

On a scheduled interval

Through event-driven projections

Through manual refresh

Recommended default: Use event-driven updates for important status changes with a manual refresh option and a visible last-refresh timestamp.

Open Question 13-02 — Performance targets

What response-time targets should apply to dashboards and searches?

Recommended default: Establish separate targets for ordinary searches, complex filtered searches, and large exports before production deployment.

Open Question 13-03 — Saved searches

Should users be able to save personal or shared searches?

Recommended default: Permit personal saved searches in Sprint 13 and defer shared organizational searches until Sprint 14 or later.

Open Question 13-04 — Search-term retention

How long should search terms and filter details remain in audit records?

Recommended default: Retain the audit event and a protected query reference while minimizing storage of sensitive free-text search terms.

Open Question 13-05 — Financial masking

Should unauthorized users see masked values such as •••• or no value at all?

Recommended default: Omit unauthorized financial fields rather than display masked values that may reveal the existence of restricted information.

Open Question 13-06 — Dashboard ownership

Should users be able to customize dashboard layouts?

Recommended default: Provide role-specific fixed layouts initially and defer personal widget customization.

Open Question 13-07 — Exception priority

How should multiple simultaneous exceptions be prioritized?

Recommended default: Prioritize safety or legal timing requirements, then overdue tasks, expiring signatures, payment blockers, approval blockers, data-quality issues, and informational warnings.

Open Question 13-08 — Expiring-document threshold

How many days before expiration should a document appear as nearing expiration?

Recommended default: Display warnings at 14 days, 7 days, and 1 day before expiration.

Open Question 13-09 — Search of historical versions

Should historical Master Sales Record and document versions appear in ordinary search results?

Recommended default: Show current versions in ordinary search and provide historical versions through an authorized version-history action.

Open Question 13-10 — Search of cancelled projects

Should cancelled projects appear in ordinary results?

Recommended default: Exclude cancelled projects by default but provide an authorized status filter to include them.

Open Question 13-11 — Search export limits

What limits should apply to search-result exports?

Recommended default: Require authorization, log every export, apply maximum row limits, and require an export reason for sensitive financial results.

Open Question 13-12 — Dashboard financial totals

Should authorized financial users see aggregate totals directly on the dashboard?

Recommended default: Provide aggregate totals only to the Sales Manager and Comptroller, with separate permission controls for each financial category.

Open Question 13-13 — Stale task handling

When a project is reopened or superseded, how should old tasks appear?

Recommended default: Mark them superseded, retain them in history, and exclude them from active-task counts unless an unresolved action remains.

Acceptance Criteria

Sprint 13 is accepted when:

Authorized users can access a role-appropriate activity dashboard.

Dashboard projects display the required project, assignment, status, task, due-date, and activity fields.

Current tasks are derived from the controlled lifecycle.

Next required actions are accurately identified.

Days in current status are calculated correctly.

Last activity date and time are displayed correctly.

Dashboard filters work individually and in combination.

Signature, installation, payment, scope, profitability, commission, payable, and reporting exceptions appear correctly.

Sales Associates see only authorized projects and their own commission information.

Technicians see only assigned or authorized projects.

Contractors see only assigned projects and required fields.

Sales Managers and Comptrollers can view authorized company-wide information.

Financial amounts are hidden from unauthorized users.

Unauthorized records do not appear in dashboard counts or results.

Users can search by all required customer, project, location, personnel, equipment, vendor, payable, document, status, and date fields.

Partial, exact, case-insensitive, normalized phone, address, project-code, and serial-number searches work as designed.

Search results are paginated or otherwise controlled for large result sets.

Search result fields vary according to the user's permissions.

Unauthorized fields are not exposed through search, sorting, filters, autocomplete, or exports.

Opening a record from search applies normal permissions.

Historical versions are handled according to the approved search policy.

Cancelled and superseded records follow the configured search behavior.
Dashboard and search activity is recorded in the tamper-evident audit log.
Permission failures are logged.
Export, print, and download actions are authorized and audited.
Dashboard data refreshes according to the configured refresh model.
Search and dashboard performance meets the approved targets.
Search results remain consistent with authoritative database records.
Financial and operational reports linked from the dashboard open only for authorized users.
Automated Sprint 13 tests pass.

15. Automated Test Scenarios

Dashboard Tests

Display a project with all required dashboard fields.
Confirm the current task for a project awaiting customer contract signature.
Confirm the current task for a project awaiting payment verification.
Confirm the current task for a project awaiting Certificate signature.
Confirm days in current status.
Confirm last activity updates after a recorded action.
Filter by project status.
Filter by Sales Associate.
Filter by Technician.
Filter by state.
Filter by payment status.
Filter by overdue tasks.
Filter by expiring documents.
Filter by negative-profit projects.
Combine multiple filters.
Confirm that unauthorized projects do not affect counts.

Role Tests

Confirm a Sales Associate sees assigned projects only.
Confirm a Sales Associate sees only their own commission information.
Confirm a Technician sees assigned projects only.
Confirm an Installation Manager sees installation-related projects.
Confirm an Accounts Payable Associate sees authorized payable records.
Confirm the Sales Manager sees all authorized project records.
Confirm the Comptroller sees all authorized financial records.
Confirm contractors see only assigned records.
Confirm unauthorized financial amounts are omitted.

Search Tests

Search by customer name.
Search by phone number in multiple formatting styles.
Search by email address.
Search by project identification code.
Search by address.
Search by serial number.
Search by brand and model.
Search by vendor.
Search by payable.
Search by document type.
Filter search by date range.
Filter search by status.
Combine search terms and filters.
Confirm partial and case-insensitive search behavior.
Confirm pagination.

Confirm empty-result behavior.

Confirm cancelled-project behavior.

Confirm historical-version behavior.

Security Tests

Attempt to search for an unauthorized customer.

Confirm no restricted record count is returned.

Attempt to sort by an unauthorized financial field.

Attempt to export unauthorized results.

Attempt to access a restricted document from search.

Confirm every permission failure is audited.

Confirm autocomplete does not expose restricted records.

Audit Tests

Access the dashboard.

Run a search.

Apply filters.

Open a result.

Export results.

Print results.

Download a document.

Confirm all events appear in the tamper-evident audit chain.

16. Sprint 13 Completion Definition

Sprint 13 is complete when:

Role-specific dashboards are operational.

Current tasks and next actions are accurately derived from workflow state.

Required operational and exception indicators are displayed.

Search covers all approved search fields.

Search respects record, field, action, and export permissions.

Financial information is protected from unauthorized exposure.

Dashboard and search results remain consistent with authoritative records.

Access, searches, exports, and failures are audit logged.

Performance and usability requirements are met.

Acceptance tests pass.

Open questions are resolved or formally carried forward.

The Sprint 13 bundle is approved and preserved unchanged.

17. Next Markdown Bundle

The next bundle is:

text

GECC-14-notifications-and-files.md

Its objective is to implement:

In-app notifications

Email notifications

Customer and project folders

Document storage references

Document metadata

File-version management

Delivery tracking

Signature-related reminders

Expiration notifications

Approval notifications

Payment-verification notifications

Commission-report distribution

Annual-report email process

Notification preferences
Failed-delivery handling
File-access permissions
File-download audit records
Customer/project document search integration
Retention and archival rules
Notification and file-management testing
===== END FILE: GECC-13-Dashboard-And-Search.md =====
=== BEGIN FILE: GECC-14-Notification-And-Files.md ===
Sprint 013 approval recorded.

GECC-14-notifications-and-files.md
GECC Sales Back Office System — Sprint 14
Bundle Metadata
Bundle name: GECC-14-notifications-and-files.md
Project: GECC Sales Back Office System
Company: Go Ecco Climate Control
Sprint: 14
Sprint name: Notifications, Email, and File Management
Baseline: `GECC-00-project-baseline.md`
Previous bundle: `GECC-13-dashboard-and-search.md`
Status: Implementation bundle
Date: 2026-09-20
Next planned bundle: GECC-15-testing-and-integrity.md
Primary database: Relational database
System of record: GECC database
Google Sheets: Export or reporting destination only

1. Sprint Objective

Implement in-app notifications, email delivery, customer and project file management, document metadata, delivery tracking, retention controls, and file-access auditing.

Sprint 14 must ensure that:

Users are notified of tasks and exceptions relevant to their roles.
Email messages are linked to the correct project, customer, document, or report.
Generated and signed documents are retained in the appropriate customer/project file.
Original and superseded document versions remain preserved.
File access follows record and role permissions.
Email delivery status is recorded.
Failed deliveries create actionable exceptions.
File downloads, emails, and document access are audited.
The annual contractor-report email process is supported.
Notification activity does not bypass the controlled workflow.

2. Sprint Scope

2.1 Included

In-app notifications
Email notifications
Notification events
Notification recipients
Notification status
Notification preferences
Customer folders
Project folders

- Document metadata
- File references
- Generated-document storage
- Signed-document storage
- Document version links
- Superseded-document handling
- Email delivery tracking
- Failed-delivery handling
- Signature-related reminders
- Expiration notifications
- Approval notifications
- Payment-verification notifications
- Commission-report distribution
- Annual contractor-report email process
- File-access permissions
- File download auditing
- Retention metadata
- File and notification search integration

2.2 Excluded

- Replacing the electronic-signature system
- Replacing the relational database
- Automatic bank notifications
- Customer self-service portal
- General marketing email
- Bulk unsolicited email
- Employee human-resources notifications
- External document collaboration
- Automatic legal-retention classification without approved policy
- Full enterprise content-management functionality

3. Notification Architecture

3.1 Notification Event

A notification event must be created from a recognized system event, including:

- Workflow task assignment
- Approval request
- Approval completion
- Approval rejection
- Signature request
- Signature reminder
- Signature expiration
- Payment verification required
- Payment verified
- Payment rejected
- Adjustment approval required
- Payable approval required
- Commission report ready for review
- Commission report approved
- Annual contractor summary ready for delivery
- Financial report ready for approval
- Financial period locked
- Exception created
- Failed email delivery
- File upload completed

File processing failed

The system must not create notifications from unsupported or unaudited event types.

3.2 Notification Record

Required fields:

Notification ID

Notification type

Event type

Recipient user ID or external recipient reference

Customer ID, if applicable

Project ID, if applicable

Document ID, if applicable

Report ID, if applicable

Related task ID, if applicable

Subject

Message body or template reference

Priority

Delivery channel

Status

Created date and time

Sent date and time

Read date and time

Expiration date, if applicable

Failure reason

Retry count

Related audit reference

3.3 Notification Channels

Initial channels:

In-app notification

Email

The system must distinguish between:

Notification created

Notification queued

Notification sent

Notification delivered

Notification opened or read

Notification failed

Notification dismissed

Notification expired

A notification may be delivered through more than one channel.

3.4 Notification Priorities

Permitted values:

LOW

`NORMAL

HIGH

URGENT

Urgent notifications may include:

Payment verification failure
Expiring signature request
Expired signature request
Closed-period adjustment issue
Security or permission exception
Failed required document delivery

4. Notification Rules

4.1 Assignment Notifications

Notify an assigned user when:

A project is assigned.
A project task is assigned.
A task is reassigned.
A task becomes overdue.
A task is cancelled or superseded.
The notification must include:

Project identification code
Customer or project reference permitted for the recipient
Task
Due date
Required action
Link to the authorized record

4.2 Approval Notifications

Notify authorized approvers when:

A Master Sales Record awaits approval.
A scope change awaits approval.
A payment adjustment awaits approval.
A refund awaits approval.
A commission report awaits review or approval.
A payable awaits approval.
A financial report awaits approval.
A formal adjustment awaits approval.
Approval notifications must not grant approval authority. They only direct the user to the authorized workflow.

4.3 Signature Notifications

Integrate with Sprint 08 for:

Signature request creation
Signature reminders
Signature delivery failure
Signature expiration
Signature completion
Required substitute signer action
The notification system must not create a second, conflicting signature request.

4.4 Payment Notifications

Notify authorized users when:

A payment is entered.
A payment requires verification.

A cashier's-check verification is rejected.
A bank wire lacks required evidence.
A payment is verified.
A payment is reversed.
A refund changes project payment status.
A project remains blocked by missing verified payment.
Customer notifications must be sent only through approved templates and only when the applicable workflow authorizes customer communication.

4.5 Commission Notifications

Notify authorized users when:

A project becomes commission-eligible.
A project is added to a commission period.
A commission report is ready for review.
A commission report is approved.
A commission report is rejected.
A commission adjustment is required.
An annual contractor summary is generated.
Annual-report delivery fails.
Contractor notifications must contain only that contractor's authorized commission information.

4.6 Financial-Report Notifications

Notify the Sales Manager and Comptroller when:

A financial period is ready for preparation.
Financial reports are ready for review.
A report is awaiting a signature.
A report is published.
A period is locked.
A formal adjustment is requested.
A ledger or reconciliation exception blocks publication.

5. Notification Preferences

Users may configure permitted preferences for:

In-app notifications

Email notifications

Reminder frequency

Non-urgent notification delivery

Daily summary notifications, if enabled

The system must not allow users to disable required notifications for:

Approval responsibilities

Payment verification responsibilities

Signature expiration actions

Financial statement approvals

Security events

Required legal or operational retention events

Notification preference changes must be audited.

6. Email Delivery

6.1 Email Message Record

Each outgoing email must retain:

Email message ID
Recipient
Sender identity or configured sender
Subject
Template ID and version
Related customer, project, document, report, or task
Attachment references
Created date and time
Queued date and time
Sent date and time
Delivery status
Failure reason
Retry count
User or system process initiating the email
Audit reference

6.2 Email Statuses

Permitted statuses:

` DRAFT
QUEUED
SENDING
SENT
DELIVERED
BOUNCED
FAILED
CANCELLED
EXPIRED

6.3 Email Templates

Templates must be versioned.

A template must include:

Template ID
Template name
Purpose
Recipient type
Subject template
Body template
Supported variables
Language, if applicable
Active status
Effective date
Retirement date
Approved by
Approval date
Version number
Templates must use controlled project and document data.

Users must not be able to insert unauthorized financial or personal information into a template through ordinary free-text fields.

6.4 Attachments

Attachments must reference stored document records rather than unmanaged local files.

The system must verify:

Recipient authorization

Document status

Document version

File availability

File integrity

Attachment size

Attachment type

A superseded or unauthorized document must not be attached automatically.

6.5 Failed Delivery

When an email fails:

Record the failure status.

Preserve the message and attachment references.

Record the failure reason.

Create an exception notification.

Create a retry task when permitted.

Avoid duplicate messages during retry.

Record every retry attempt.

Notify the responsible authorized user when the message is operationally required.

7. File and Folder Architecture

7.1 Customer Folder

Each customer may have a customer-level file area containing:

Customer agreements

General correspondence

Customer identity or contact documents

Customer-level reports

Archived files

Approved supporting records

Access must follow customer-level authorization.

7.2 Project Folder

Each project must have a project-level file area containing:

Master Sales Record versions

Invoice versions

Contract versions

Certificate versions

Signed-document certificates

Payment evidence

Adjustment documents

Scope-change documents

Equipment records or supporting files

Commission-report references

Project correspondence

Financial-supporting records

Audit-linked file references

Project documents must remain associated with the customer and project.

7.3 File Record

Required fields:

File ID
File name
File type
MIME type
File size
Storage reference
Checksum or integrity hash
Customer ID
Project ID
Document ID, if applicable
Document version
File category
Source system
Uploaded by
Upload date and time
Status
Supersedes file ID, if applicable
Retention classification
Retention start date
Retention end date, if applicable
Access classification
Notes

7.4 File Statuses

Permitted statuses:

UPLOADING
AVAILABLE
PROCESSING
PROCESSING_FAILED
SUPERSEDED
ARCHIVED
QUARANTINED
CANCELLED

A file that is superseded or archived remains historically available to authorized users.

7.5 File Categories

Initial categories include:

Master Sales Record
Invoice
Contract
Certificate
Signature audit certificate
Payment proof
Adjustment support
Payable support
Commission report
Annual contractor summary
Financial report